

UNIVERSITÉ CATHOLIQUE DE LOUVAIN
ÉCOLE POLYTECHNIQUE DE LOUVAIN



LINFO1115

REASONING ABOUT A HIGHLY CONNECTED WORLD : GRAPH THEORY,
GAME THEORY, AND NETWORKS

Project – The Railway Network Analysis

Auteur :

Joe Ziadeh, 6854-23-00

joe.ziadeh@student.uclouvain.be

Kevin Tekli, 1676-22-00

kevin.tekli@student.uclouvain.be

Task 0 – Dataset Overview

Results

Metric	Value
Distinct nodes	47234
Edges	54644
Is the graph connected?	No
Number of connected components	52

TABLE 1 – Basic statistics of the European railway graph (`total.csv`).

Only path from Ahlbeck_Grenze to Peenemunde : Ahlbeck_Grenze → Seebad_Ahlbeck → Ahlbeck_Ostseetherme → StrUeb6768_6773_6400 → Heringsdorf__Neuhof → Bansin_Seebad → Schmollensee → Uckeritz → Stubbenfelde → Kolpinsee → Koserow → Zempin → Zinnowitz → StrUeb6773_6774_8168 → Trassenmoor → Karlshagen → Peenemunde **Length :** 38.6714 km.

Shortest path from Portarlinton_Junction to Foyens_Junction : All 11 simple paths enumerated ; minimum length = 125.4776 km.

Time Complexity

Node/edge count : $\mathcal{O}(E)$.

Components (BFS) : $\mathcal{O}(V + E)$.

Only path (Dijkstra) : $\mathcal{O}((V + E) \log V)$.

11-path DFS : $\mathcal{O}(P \cdot L)$, negligible on the small Irish sub-graph.

Task 1 – Graph Structure

Results

Metric	Value
Average degree	2.313757
Number of bridges	9012
Number of local bridges	48877

TABLE 2 – Structural metrics.

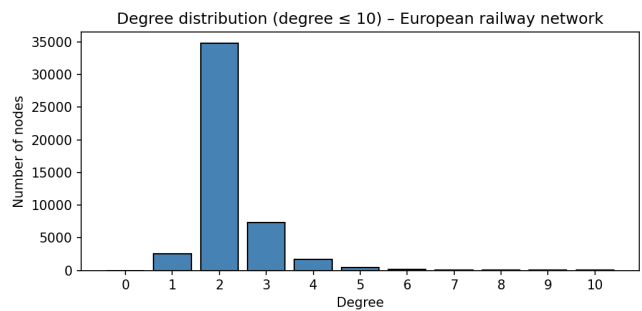


FIGURE 1 – Degree distribution (degrees ≤ 10).

The average degree of ≈ 2.313757 confirms a near-linear topology : most stations lie on a single line with degree 2, while junctions and terminals are rare.

The large bridge count (9012) shows the network has almost no redundant loops – any single track failure can disconnect a region. Local bridges (48877) mark the sole inter-regional links between otherwise disjoint sub-networks.

Time Complexity

Average degree : $\mathcal{O}(E)$.

Bridges (iterative Tarjan) : $\mathcal{O}(V + E)$.

Local bridges : $\mathcal{O}(E \cdot \bar{d})$.

Task 2 – Clustering and Balance

Results

Station	CC
Berlin_Westhafen	0.333333
Krakow_Gowny	0.190476
Amsterdam_Transf._Aansl.	0.100000
ROMA_TERMINI	0.333333

TABLE 3 – Local clustering coefficients.

Metric	Value
Total triangles	2353
Balanced triangles	1245
Unbalanced triangles	1108
GCC	0.011332

TABLE 4 – Triangles (sign : $d < 1 \text{ km} = +$, $d \geq 1 \text{ km} = -$) and GCC.

All four CCs are close to zero, as expected : railway lines minimise cost and rarely form triangular loops.

The low GCC confirms a globally tree-like topology. Most triangles are balanced (0 or 2 negative edges) : short positive edges cluster nearby infrastructure nodes, while long negative edges separate them, naturally satisfying balance.

Time Complexity

Local CC : $\mathcal{O}(k^2)$ per node.

Triangles & balance : $\mathcal{O}(E\sqrt{E})$.

GCC : $\mathcal{O}(V \cdot \bar{d}^2)$.

Task 3 – Betweenness Centrality (Belgian Network)

Results

Metric	Value
Top station	Antwerpen-Berchem
Raw BC score	215749
Normalised score	0.245225

TABLE 5 – Betweenness centrality (belgium.csv).

Normalised by $C = \frac{(V-1)(V-2)}{2}$

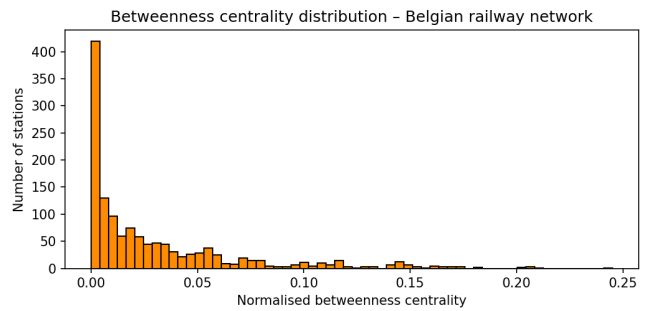


FIGURE 2 – Normalised BC distribution.

The most shortest paths concentrates between the stations Antwerpen & Berchem, acting as the primary bottleneck of the Belgian network ; its disruption would raise travel costs across many origin-destination pairs. The distribution is strongly right-skewed : nearly all stations have near-zero BC while a handful of hubs carry the bulk of the routing load.

Time Complexity

Brandes + Dijkstra : $\mathcal{O}(V \cdot (V + E) \log V)$.

Task 4 – Station Score and Network Improvement (Belgian Network)

Results

$$\text{score}(n) = \deg(n) \times \sum_{i \in \mathcal{N}(n)} \frac{\deg(i)}{d(n, i)} \times \sum_{j \in \mathcal{N}(i) \setminus \{n\}} \deg(j) \cdot d(i, j)$$

Rk	Station (highest)	Score
1	Forest-Midi-Voies_920/926	141837.33
2	Ruisbroek-Bundel-Inrit	40580.60
3	VERB.Polymers.II	37728.92
4	Forest-Midi_Vorst-Zuid	35765.77
5	Y.Schijn	9506.72

TABLE 6 – Top-5 stations by score.

Rk	Station (lowest)	Score
1	Lanaken	0.0000
2	Mariembourg	0.4016
3	Racc.Engis-Carmeuse	1.0839
4	Quievrain	1.3936
5	Meerhout-E.C.M.	1.4499

TABLE 7 – Bottom-5 stations by score.

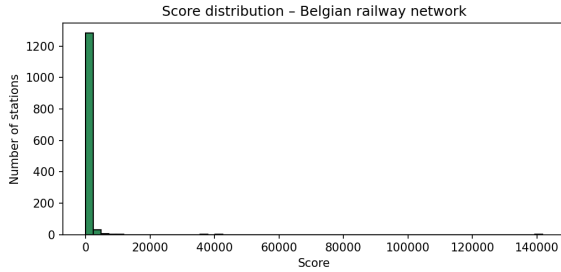


FIGURE 3 – Score distribution.

$$\text{criterion}(u, v) = \frac{\Delta \text{score}}{\text{score}(u) + \text{score}(v)} \times d(u, v) \times 100$$

Rk	Edge (u, v)	Crit.
1	Ieper – Poperinge	3092.15
2	Couvin – Mariembourg	1806.00
3	Eeklo – Waarschoot	894.38
4	Sterpenich-Fr. – Y.Autelbas	862.93
5	Kleinbettingen_fr. – Y.Autelbas	862.93

TABLE 8 – Top-5 edges for midpoint insertion.

Interpretation

High-scoring stations are dense urban yards or major junctions with many close neighbours that themselves have wide reach ; low-scoring ones are isolated branch endpoints. The score distribution is strongly right-skewed, with a small elite of hubs dominating. The criterion favours long edges between important stations where a midpoint insertion yields the greatest network-wide gain relative to baseline connectivity.

Time Complexity

All scores : $\mathcal{O}(V \cdot \bar{d}^2)$.

Criterion per edge : $\mathcal{O}(\bar{d}^2)$ after caching scores.